

DataSense: A Real-Time Sensor-Based Benchmark Dataset for Attack Analysis in IIoT with Multi-Objective Feature Selection

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raw_files:

The `raw_files` directory contains raw captured data from the testbed, including both network traffic (PCAP files) and sensor data (JSON files). It includes benign and attack scenarios, where each set of files is time-synchronized so that network packets and sensor readings can be accurately matched using timestamps.

The **raw_files** directory contains two subdirectories:

- **attack_data**

- Raw captured network traffic in PCAP format during different attack scenarios.
- Sensor data collected simultaneously from the testbed devices in JSON format.
- Both PCAP and JSON files are time-aligned, allowing packets and sensor events to be correlated through timestamps.
- Each attack scenario targets one or more devices depending on the nature of the attack.
- File names follow the pattern:

`{data_type}{attack_category}{attack_name}_{target_device}`

- Example:
 - `attack_web_backdoor-upload_edge1`
 - Category: web attacks
 - Attack name: backdoor-upload
 - Target device: edge1
- Both pcap and .json files follow the same pattern and for each pcap a json file with the same name exists that show the sensor data for that specific scenario.

- **benign_data**

- Raw captured network traffic in PCAP format during normal (benign) operation of the testbed.
- Sensor data collected simultaneously from the testbed devices in JSON format.
- Files are also time-synchronized, providing a consistent baseline for comparison against attack scenarios.